

KEZHOU LU

Assistant Professor

Department of Earth, Ocean and Atmospheric Science

Florida State University

☎ (+1) 510-820-7974

✉ melodylu@fsu.edu

🌐 Personal Website

🐙 Github in LinkedIn

Education

- 2018–2023 **Ph.D., Ocean & Climate Dynamics**, Georgia Institute of Technology, Atlanta, GA, USA.
Minor: Applied Mathematics
- 2017–2018 **M.S., Environmental Engineering**, University of California, Berkeley, Berkeley, CA, USA.
- 2013–2017 **B.S., Environmental Science**, Tongji University, Shanghai, China.

Employment

- Aug 2026–
present **Assistant Professor**, Florida State University, Tallahassee, FL, USA.
- Sep 2024–
Jul 2026 **Postdoctoral Scientist**, University of California, Los Angeles, Los Angeles, CA, USA.
- Jan–May 2024 **Lecturer**, Spelman College, Atlanta, GA, USA.

Professional Experience

Postdoctoral Scientist — University of California, Los Angeles

- Oct 2025–
present **Neural Hybrid Residual XRO Models for ENSO Forecasting.**
- Collaborated on the development of neural residual ODE model for ENSO forecasting that combine physics-based Extended Recharge Oscillator (XRO) dynamics with neural components.
- Nov 2024–
present **Land Precipitation Response to CO₂ and Its Impacts on Subtropical Pacific SST.**
- Conducted a 120-member abrupt-4xCO₂ large ensemble with CESM to isolate the mechanisms driving rapid cooling in the subtropical South Pacific.
 - By designing a set of novel diabatic heating perturbation experiments that prescribe tropical South America heating anomalies in an atmosphere–slab ocean model, we reproduced the subtropical Pacific cooling and low-level wind responses seen in the large ensemble.
- Sep 2024–Jun
2025 **Dynamical Constraints on Mid- and High-Latitude Temperature Distribution.**
- Reconstructed low-level atmospheric temperature distributions from a Lagrangian perspective.
 - Designed and conducted **a hierarchy of model experiments**—from Held–Suarez to fully coupled GCMs—to quantify the relative roles of thermodynamic and dynamical processes in shaping temperature responses to anthropogenic forcing.
- Jan–Jun 2025 **Land–Atmosphere Feedbacks and Their Impact on Surface Temperature Variance.**
- Developed a **theoretical framework** to quantify the influence of land–atmosphere coupling on surface temperature variance over mid- and high-latitude continents.
 - Validated the framework using numerical experiments with the CESM2 model.

Lecturer — Spelman College

- Jan–May 2024 **Instructor of an undergraduate course "Air and Atmosphere".**

Graduate Research Assistant — Georgia Institute of Technology

- Jan 2022–Nov
2023 **Diverging Hydrological Sensitivity Among Tropical Basins.**
- Conducted diabatic heating perturbation experiments using CESM1 and a stationary wave model to evaluate large-scale atmospheric circulation responses.
 - Demonstrated the global climate impact of inter-basin differences in tropical precipitation sensitivity.

- Jan 2021–Mar 2023 **Origin of Model Uncertainty in the North Pacific Subtropical High.**
- Identified tropical precipitation uncertainty as a key contributor to inter-model spread in projected changes of the North Pacific Subtropical High.
 - Established and validated the causal relationship using CESM1 and a stationary wave model.
- Sep 2018–Dec 2020 **Mechanisms of Fast Walker Circulation Responses to CO₂ Forcing.**
- Identified divergent fast Walker Circulation responses among CMIP5 models under abrupt-4×CO₂ forcing, and confirmed that the inter-model spread is externally forced using a CESM1 large ensemble.
 - Demonstrated that these inter-model differences are driven by differences in models' mean-state tropical Pacific air–sea coupling strength.

Publications

(*student advisee)

Peer-reviewed Journal Articles

- Jun 2026 Bowen Ge, Gang Chen, and **Kezhou Lu**. Mechanisms of southern hemisphere midlatitude surface air temperature variability under climate warming in the cesm2 large ensemble (lens2). *Journal of Climate*, Jun 2026. DOI: <https://doi.org/10.1175/JCLI-D-25-0435.1>.
- Feb 2026 **Kezhou Lu**, Gang Chen, Bowen Ge, Rong Fu, Weiming Ma, and Hailong Wang. The role of land-atmosphere feedback on midlatitude wintertime surface temperature variance. *Geophysical Research Letters*, Feb 2026. DOI: <https://doi.org/10.1029/2025GL120713>.
- Feb 2026 Bai Bin et al (including **Kezhou Lu**). Hygroscopicity and cloud condensation nuclei (ccn) activity of fresh and aged biomass burning particles. *ACS ES&T Air*, Feb 2026. DOI: <https://doi.org/10.1021/acsestair.5c00331>.
- Apr 2026 Yue Dong, **Kezhou Lu**, Yen-Ting Hwang, Ruei-Jia Hu, Paulo Ceppi, Breul Philipp, Lettie Roach, and Clara Deser. Tropical climate impacts of the southern ocean underestimated by mean-state biases. *Science Advances*, Apr 2026.
- Sep 2024 Jie He, Yi Deng, Boniface Fosu, Yen-Heng Lin, and **Kezhou Lu**. Constraining regional hydrological sensitivity over tropical oceans. *Geophysical Research Letters*, volume 4, Sep 2024. DOI: <https://doi.org/10.1029/2024GL109374>.
- Apr 2024 Jie He, **Kezhou Lu**, Boniface Fosu, and Fueglistaler Stephan. Diverging hydrological sensitivity among tropical basins. *Nature Climate Change*, volume 14, Apr 2024. DOI: <https://doi.org/10.1038/s41558-024-01982-8>.
- Nov 2023 **Kezhou Lu**, Jie He, and Isla Simpson. Origins of uncertainty in the response of the summer North Pacific Subtropical High to CO₂ forcing. *Geophysical Research Letters*, volume 50, Nov 2023. DOI: <https://doi.org/10.1029/2023GL105042>.
- Nov 2021 **Kezhou Lu**, Jie He, Boniface Fosu, and Maria Rugenstein. Mechanisms of fast Walker circulation responses to CO₂ forcing. *Geophysical Research Letters*, volume 48, Nov 2021. DOI: <https://doi.org/10.1029/2021GL095708>.

Submitted Manuscripts

- Dec 2025 **Kezhou Lu**, Yue Dong, Gang Chen, Stephan Fueglistaler, and Clara Deser, Subtropical Pacific cooling driven by land rainfall response to CO₂ Forcing (under review at *Nature Geoscience*).
- Feb 2026 Fred Xu, **Kezhou Lu**, Dmitri Kondrashov, Yizhou Sun, Gang Chen, Neural Hybrid Residual XRO Models for ENSO Forecasting (under review at *KDD 2026 Conference*).

Manuscripts in Preparation

- Dec 2025 **Kezhou Lu**, Gang Chen, and Bowen Ge, Dynamical constraints on mid- and high-latitude temperature distribution.

Other Work

- Nov 2023 **Kezhou Lu**, Uncertainties in Projections of Tropical Precipitation and Atmospheric Circulation and Their Remote Impacts. *Georgia Institute of Technology*. Ph.D. Dissertation.

- May 2022 *Enrique Saldivar, **Kezhou Lu**, and Zack Handlos, Investigating the relationship between the El Niño Southern Oscillation and west Pacific jet vertical superposition events. *Honors Undergraduate Thesis, San Jose State University*.

Selected Presentations

Invited Talks

- Jun 2026 Critical impact of land-atmosphere feedbacks on midlatitude wintertime surface temperature variance. In *AGU Atmospheric Science Section Early Career Seminar*. AGU.
- Feb 2026 Large-scale atmospheric circulation across the earth system: From global patterns to regional extremes. In *MET Seminar*. Florida State University.
- Aug 2025 Origins of uncertainty in the response of the north Pacific Subtropical High. In *Geophysical Institute Science FRI Seminar*. University of Alaska, Fairbanks.
- Oct 2024 Origins of uncertainty in the response of the north Pacific Subtropical High. In *AOS Department Seminar*. University of California, Los Angeles.
- May 2024 Diverging hydrological sensitivity among tropical basins. In *Group Seminar*. Princeton University.
- Mar 2024 Origins of uncertainty in the response of the north Pacific Subtropical High (as **Keynote Speaker**). In *Graduate Student Symposium*. Georgia Institute of Technology.
- Nov 2023 Mechanisms of fast Walker circulation responses to CO_2 forcings. In *Equilibrium Climate Sensitivity and Cloud Feedback Virtual Symposium*.
- Feb 2022 Mechanisms of fast Walker circulation responses to CO_2 forcings. In *CESM Climate Variability and Change Group Meeting*. NCAR.

Contributed Talks and Posters

- Dec 2025 AGU Annual Meeting, *Talks*.
- Sep 2025 CalGFD 2025, *Talk*.
- Jun 2025 CESM Annual Workshop, *Talk*.
- Dec 2024 AGU Annual Meeting, *Talk*.
- Jun 2023 CESM Annual Workshop, *Talk*.
- Jan 2023 AMS Annual Meeting, *Talk*.
- Oct 2022 Akio Arakawa Symposium, *Poster*.
- Jan 2022 AMS Annual Meeting, *Talk*.
- Dec 2020 AGU Annual Meeting, *Virtual Poster*.
- Dec 2019 AGU Annual Meeting, *Poster*.
- Jul 2019 Large Ensemble Workshop at NCAR, *Poster*.

Fellowships & Awards

Awards

- May 2023 **Outstanding Oral Presentation** at Graduate Symposium, Georgia Institute of Technology.
- Dec 2022 **Student Recognition of Excellence in Teaching: Class of 1934 CIOS Honor Roll** with Dr. Sven Simon for class "Earth System Modeling", Georgia Institute of Technology.
- May 2024 **Outstanding Teaching Assistant**, Georgia Institute of Technology

Fellowships & Travel Grants

- Jun 2025 **Early-career Scientist Travel Grant** to attend CESM annual workshop, \$2000, NCAR.
- Sep 2023 **Graduate Student Travel Scholarship**, \$1500, Georgia Institute of Technology.
- Jul 2023 **ForceSMIP Hackathon Scholarship**, \$2000, NCAR.
- Oct 2022 **Akio Arakawa Symposium Scholarship**, \$2000, University of California, Los Angeles.

- 2021–2022 **Douglas Davis Fellowship in Atmospheric Sciences** , \$2500, Georgia Institute of Technology.
 2017–2018 **Room & Board Scholarship** , \$5000, University of California, Berkeley.
 Jul 2017 **Gu-Guowei Scholarship for Excellent Undergraduate Thesis** , ¥5000, Tongji University.

Teaching & Mentoring

Teaching

- Jan 2024–May 2024 **Lecturer**, *Spelman College*.
 ○ Developed and taught a 4-credit introductory course: "*Air and Atmosphere*".
 ○ Designed hands-on labs, assignments and exams to support student learning.
- Aug 2023–Dec 2023 **Guest Lecturer**, *Georgia Institute of Technology*.
 with Dr. Jie He for "*Large-Scale Atmospheric Circulation*"
 ○ Delivered tutorials guiding students on how to run and analyze baroclinic stationary wave model.
- Jan 2024–May 2024 **Guest Lecturer**, *Georgia Institute of Technology*.
 with Dr. Annalisa Bracco for "*Climate Modeling*"
 ○ Delivered tutorials guiding students on how to run and analyze CESM model.
- 2019–2021 **Head Teaching Assistant**, *Georgia Institute of Technology*.
 with Dr. Samantha Wilson for "*Introduction to Environmental Sciences Lab*"
- Aug 2021–Dec 2021 **Teaching Assistant**, *Georgia Institute of Technology*.
 with Dr. Sven Simon for "*Earth System Modeling*"

Mentoring

- May 2026–present **Tyler Do**, *undergraduate student at University of California, Los Angeles*.
 co-advised with Dr. Gang Chen
- Aug 2025–present **Siofra Linden**, *undergraduate student at University of California, Los Angeles*.
 co-advised with Dr. Gang Chen
- Aug 2025–present **Anna Coomans**, *master student at University of California, Los Angeles*.
 co-advised with Dr. Gang Chen
- Sep 2024–present **Rui Zhang**, *master student at University of California, Los Angeles*.
 co-advised with Dr. Gang Chen
- May 2022–Aug 2023 **Enrique Saldivar**, *undergraduate student at San Jose State University, REU*.
 co-advised with Dr. Zack Handlos
- May–Jul 2022 **Abigail McDonnell**, *undergraduate student at University of Illinois Urbana-Champaign, REU*.
- May–Jul 2019 **Nidhi Desai**, *undergraduate student at Cornell University, REU*.

Academic Services

Journal Referee

Journal of Climate (1), Science Advances (2), Weather and Climate Dynamics (1), Journal of Geophysical Research: Atmosphere (11), Advances in Atmospheric Sciences (1), Scientific Reports (1), npj Climate and Atmospheric Science (1), Geophysical Research Letters (1)

Services

- Dec 2025 Co-chair of AGU Atmospheric Physics, Dynamics, and Climate Session
- Dec 2025 Co-chair of AGU The Dynamics of Large-Scale Atmospheric Circulation in Present and Future Climates: Jet Streams, Storm Tracks, Stationary Waves, and Monsoons Session
- Dec 2024 Judge of AGU Outstanding Student Presentation Award
- 2020–2021 Chair of Graduate Student Symposium, Georgia Institute of Technology

Outreach

- May 2025 Volunteer of The Weather & Climate Livestream: 100 Hours to Save America's Forecasts

Skills

- Mathematics Statistics, machine learning, numerical solution to PDEs
- Modeling Operating and modifying general circulation models (e.g., CESM), implementing simple models (e.g., baroclinic stationary wave models)
- Programming Python, Fortran, PyTorch, Matlab, R, C#, NCL, CDO, Latex